



JAIPUR NATIONAL UNIVERSITY

End/Back Semester Examination May, 2024

Name of School:	School of Engineering and Technology
Name of Programme / Course:	B.Tech. (Biomedical) / B.Tech. (Biotech) / B.Tech. (Chemical) / B.Tech. (Civill) / B.Tech. (Mechanical) / B.Tech. (CSE) / B.Tech. (CSE-AIML) / B.Tech. (CSE-Cyber Security) / B.Tech. (Foodtech) / B.Tech. (Electrical)
Semester/Year:	II Sem.
Paper Code:	BTEEMCO202T22
Title of Paper:	Engineering Mechanics
Time: 3 hrs.	Max. Marks: 60

INSTRUCTIONS / NOTES FOR THE EXAMINEE

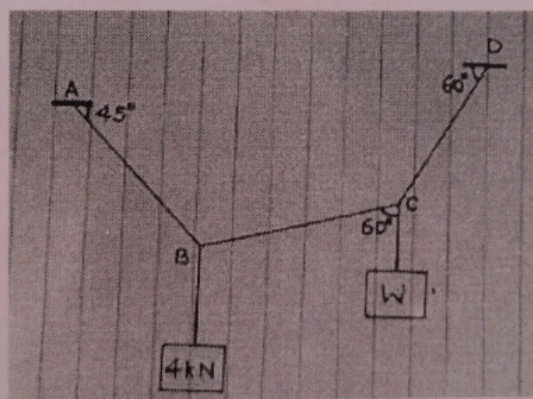
- (i) Attempt any 5 questions from Section A and any 4 questions from Section B.
- (ii) Draw neat diagram wherever required.
- (iii) Calculator is allowed.

Question Number	SECTION-A	Marks Allotted
Q1	Define the term 'Force System' also explain all the types of force system.	4
Q2	What do you understand by two force and three force principle? Explain with the help of an example.	4
Q3	Explain the following terms: (i) Static friction (ii) Limiting friction (iii) Kinetic friction (iv) Laws of friction	4

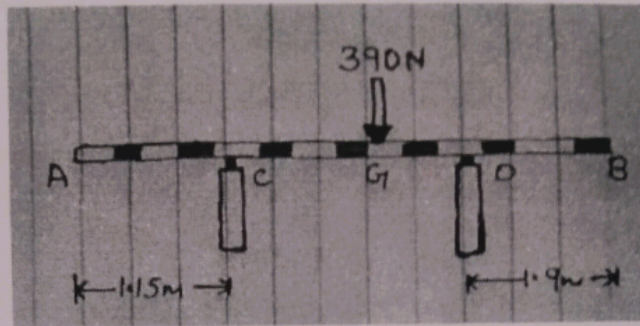
- Q4 What are the Laws of friction? Explain any 5 law. 4
- Q5 Drive and explain the following terms: 4
 (i) Time of flight (ii) Horizontal Range
- Q6 Drive and explain the Law of Conservation of Energy 4
- Q7 Derive the equation for Work – Energy relation. 4

SECTION-B

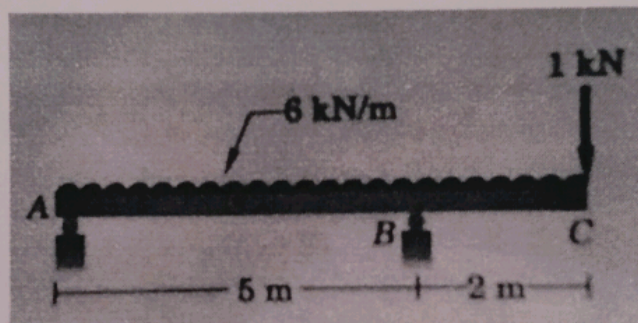
- Q1 A string is subjected to the forces 4kN and W as shown in fig. 10
 Determine the magnitude of W and the tensions in the various portion of the string.



- Q2 A uniform plank 5m long and 390N weight is supported by two supports; one is 1.15m from one end and other 1.9m from the other end. Determine the reaction of the two supports. 10
 Shyam weighing 260N walk on it starting from the second supports towards right, find how far it is safe for him to walk. Determine the reactions of the supports when he start to walk from safest point.



- Q3 A beam AC of 7m length is loaded and supported as shown in fig. Find the reaction at the support using the principle of virtual work. 10



- Q4 A ladder of length 20m rests on a floor making an angle 45° with the wall. The coefficient of friction between the ladder and the wall is 0.4 and that between ladder and the floor is 0.6. If a man, whose weight is one half of that of the ladder, ascends it, how high will it be when the ladder slips? 10

- Q5 A stone of 900 N is fastened to a light rod of length 3m and carried by Rahul and Rohit on their shoulders. Rohit being weak and cannot carry more than 150 N weight. Find out the position of stone fastened to the rod from its one end. 10

- Q6 A stone is projected with such an angle with horizontal, the range is 3 times the greatest height attained by the body. 10
- (a) Find the angle of projection

- (b) If with this angle range is 360m, find necessary velocity for the projection.
- (c) Time of flight